

Looker Developer Guide

Power BI -> LookML build playbook for developers

Source PBIX: Human Resources Sample PBIX.pbix

LookML model: human_resources_sample_pbix

Primary fact / explore: FactTable

Approach: deterministic Phase 2 (inventory-driven)

This PDF: full instructions, checks, suggestions, examples, and gaps

Inside this guide

- * Section 4: every PBI object type with step-by-step build instructions (one topic per page)
- * Section 5: concrete build plan for THIS PBIX (views, joins, measures, M)
- * Section 6: prioritized GAP list with actions
- * Sections 7-8: validation checklist and acceptance criteria

1. How to use this guide

- * Read Best practices first - mandatory.
- * Complete Prerequisites before editing LookML.
- * Read each Object equivalence page fully (Build + Check + Suggestions + Example).
- * Execute Build plan for this PBIX.
- * Close HIGH gaps (or get written sign-off).
- * Sign off using Validation and acceptance.

2. Best practices (mandatory)

Primary key on every view

Put primary_key: yes on the unique key. Required for symmetric aggregates.

Explicit join relationship

Always set relationship: many_to_one (or the correct cardinality). Never omit it.

Measures reference dimensions

Prefer sql: \${dimension} over raw \${TABLE}.col when the dimension exists.

Warehouse owns ETL

Power Query M is NOT LookML. Rebuild transforms in dbt/Dataform/SQL first.

Granular includes

Prefer include per view file (looker-skills modeling guidelines).

Explore descriptions

Every explore needs label + description.

No silent M:M / inactive joins

Review fan-out risk; use bridge tables or aliased joins intentionally.

No KPI claims without tests

Complex DAX TODOs are unfinished until side-by-side parity passes.

3. Prerequisites checklist

- * ☐ Looker project + Git branch ready
- * ☐ Admin connection created
- * ☐ Warehouse tables exist (or creation tickets filed)
- * ☐ Phase 1 inventory is for THIS PBIX
- * ☐ LOOKML_PROJECT.zip imported
- * ☐ SQL dialect known (BigQuery SAFE_DIVIDE vs Snowflake DIV0)

Do not start LookML polish until connection and warehouse tables are real. Placeholder
YOUR_LOOKER_CONNECTION / YOUR_PROJECT.YOUR_DATASET will fail validation.

4.1 Object equivalence

Table (business) -> view (.view.lkml)

What it means

Create views/<name>.view.lkml with ``view: <name> { sql_table_name: ... }``. Every view needs a `primary_key` dimension (Looker skills / symmetric aggregates).

How to build (follow these steps)

* 1. Create views/<name>.view.lkml with ``view: <name> { sql_table_name: ... }``. Every view needs a `primary_key` dimension (Looker skills / symmetric aggregates).

What to check

Suggestions / best practice

References

* <https://cloud.google.com/looker/docs/reference/param-view-view>

* <https://github.com/looker-open-source/looker-skills/blob/main/skills/lookml-view/SKILL.md>

4.2 Object equivalence

Column -> dimension (or dimension_group for dates)

What it means

Add ``dimension: field { type: ... sql: ${TABLE}.col ;; }``. Date/time columns -> ``dimension_group: ... { type: time timeframes: [raw, date, week, month, quarter, year] }``.

How to build (follow these steps)

* 1. Add ``dimension: field { type: ... sql: ${TABLE}.col ;; }``. Date/time columns -> ``dimension_group: ... { type: time timeframes: [raw, date, week, month, quarter, year] }``.

What to check

Suggestions / best practice

References

- * <https://cloud.google.com/looker/docs/reference/param-field-dimension>
- * <https://cloud.google.com/looker/docs/reference/param-field-dimension-group>

4.3 Object equivalence

Measure (DAX) -> measure

What it means

Map SUM/AVERAGE/COUNT/DISTINCTCOUNT to type: sum|average|count|count_distinct. Ratios -> type: number

+ SAFE_DIVIDE. CALCULATE/time-intel -> filters, period patterns, or warehouse + TODO.

How to build (follow these steps)

* 1. Map SUM/AVERAGE/COUNT/DISTINCTCOUNT to type: sum|average|count|count_distinct. Ratios -> type: number + SAFE_DIVIDE. CALCULATE/time-intel -> filters, period patterns, or warehouse + TODO.

What to check

Suggestions / best practice

References

* <https://cloud.google.com/looker/docs/reference/param-field-measure>

* <https://cloud.google.com/looker/docs/reference/field-reference>

4.4 Object equivalence

Relationship (M:1 From->To) -> explore join (relationship: many_to_one)

What it means

In the model file, `explore: fact { join: dim { type: left\_outer relationship: many\_to\_one sql\_on: \${fact.fk} = \${dim.pk} ;; } }`. Always set relationship explicitly (looker-skills).

How to build (follow these steps)

* 1. In the model file, `explore: fact { join: dim { type: left\_outer relationship: many\_to\_one sql\_on: \${fact.fk} = \${dim.pk} ;; } }`. Always set relationship explicitly (looker-skills).

What to check

Suggestions / best practice

References

* <https://cloud.google.com/looker/docs/reference/param-explore-join>

* <https://github.com/looker-open-source/looker-skills/blob/main/skills/lookml-explore/SKILL.md>

4.5 Object equivalence

Calculated column (DAX) -> dimension (prefer warehouse column)

What it means

Prefer materializing in warehouse SQL, then expose as dimension. Simple row expressions may use LookML sql;; complex DAX stays as migration TODO.

How to build (follow these steps)

* 1. Prefer materializing in warehouse SQL, then expose as dimension. Simple row expressions may use LookML sql;; complex DAX stays as migration TODO.

What to check

Suggestions / best practice

References

* <https://cloud.google.com/looker/docs/lookml-terms-and-concepts>

4.6 Object equivalence

Calculated table -> view (sql_table_name or derived_table)

What it means

If the table is seeded/small, warehouse seed + standard view. Otherwise SQL derived table / NDT per lookml-view derived_table guidance.

How to build (follow these steps)

* 1. If the table is seeded/small, warehouse seed + standard view. Otherwise SQL derived table / NDT per lookml-view derived_table guidance.

What to check

Suggestions / best practice

References

* <https://github.com/looker-open-source/looker-skills/blob/main/skills/lookml-view/SKILL.md>

4.7 Object equivalence

Power Query M -> Warehouse / ETL (not LookML)

What it means

Rebuild M transforms in the warehouse (dbt/Dataform/SQL). LookML only points sql_table_name at the finished table.

How to build (follow these steps)

* 1. Rebuild M transforms in the warehouse (dbt/Dataform/SQL). LookML only points sql_table_name at the finished table.

What to check

Suggestions / best practice

References

* <https://cloud.google.com/looker/docs/lookml-terms-and-concepts>

4.8 Object equivalence

Hierarchy -> drill_fields / sets / dimension_group timeframes

What it means

Date hierarchies -> dimension_group timeframes. Attribute hierarchies -> drill_fields: [year, quarter, month, day] or sets.

How to build (follow these steps)

* 1. Date hierarchies -> dimension_group timeframes. Attribute hierarchies -> drill_fields: [year, quarter, month, day] or sets.

What to check

Suggestions / best practice

References

* <https://cloud.google.com/looker/docs/reference/param-field-drill-fields>

4.9 Object equivalence

RLS / OLS -> access_grant / access_filter / required_access_grants

What it means

Map roles to Looker user attributes + access_filter on explores, or access_grant on fields.

How to build (follow these steps)

* 1. Map roles to Looker user attributes + access_filter on explores, or access_grant on fields.

What to check

Suggestions / best practice

References

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<https://github.com/looker-open-source/looker-skills/blob/main/skills/lookml-access-grants/SKILL.md>

4.10 Object equivalence

LocalDateTable_* / DataTableTemplate_* -> Skip (use business date + dimension_group)

What it means

Do not migrate auto-date tables. Use the business date column with Looker timeframes.

How to build (follow these steps)

* 1. Do not migrate auto-date tables. Use the business date column with Looker timeframes.

What to check

Suggestions / best practice

References

* <https://cloud.google.com/looker/docs/reference/param-field-dimension-group>

4.11 Object equivalence

Model / Dataset -> model (.model.lkml) + connection

What it means

One model file: connection, includes, datagroup, explore(s). Prefer granular includes over wildcards (lookml-modeling-guidelines).

How to build (follow these steps)

* 1. One model file: connection, includes, datagroup, explore(s). Prefer granular includes over wildcards (lookml-modeling-guidelines).

What to check

Suggestions / best practice

References

* <https://cloud.google.com/looker/docs/lookml-terms-and-concepts>

<https://github.com/looker-open-source/looker-skills/blob/main/skills/lookml-modeling-guidelines/SK>

* ILL.md

5. Build plan for this PBIX

Step A - Import LookML ZIP

- * Unzip LOOKML_PROJECT.zip into the Looker project.
- * Open models/<model>.model.lkml and set connection.
- * Run LookML Validator (expect table errors until Step B).

Step B - Point each view at the warehouse

For each view: set sql_table_name, confirm PK and columns exist.

View `agegroup` (Power BI: AgeGroup)

File: views/agegroup.view.lkml | columns: 2 | PK guess: AgeGroupID | date cols: none

- * Confirm primary_key matches warehouse uniqueness.
- * Map dates to dimension_group timeframes.

View `bu` (Power BI: BU)

File: views/bu.view.lkml | columns: 4 | PK guess: BU | date cols: none

- * Confirm primary_key matches warehouse uniqueness.
- * Map dates to dimension_group timeframes.

View `date` (Power BI: Date)

File: views/date.view.lkml | columns: 12 | PK guess: Date | date cols: Date, MonthStartDate, MonthEndDate

- * Confirm primary_key matches warehouse uniqueness.
- * Map dates to dimension_group timeframes.

View `employee` (Power BI: Employee)

File: views/employee.view.lkml | columns: 16 | PK guess: EmplID | date cols: date, TermDate, HireDate

- * Confirm primary_key matches warehouse uniqueness.
- * Map dates to dimension_group timeframes.

View `ethnicity` (Power BI: Ethnicity)

File: views/ethnicity.view.lkml | columns: 2 | PK guess: Ethnic Group | date cols: none

- * Confirm primary_key matches warehouse uniqueness.
- * Map dates to dimension_group timeframes.

View `fp` (Power BI: FP)

File: views/fp.view.lkml | columns: 2 | PK guess: FP | date cols: none

- * Confirm primary_key matches warehouse uniqueness.
- * Map dates to dimension_group timeframes.

View `gender` (Power BI: Gender)

File: views/gender.view.lkml | columns: 3 | PK guess: ID | date cols: none

- * Confirm primary_key matches warehouse uniqueness.
- * Map dates to dimension_group timeframes.

View `paytype` (Power BI: PayType)

File: views/paytype.view.lkml | columns: 2 | PK guess: PayTypeID | date cols: none

- * Confirm primary_key matches warehouse uniqueness.
- * Map dates to dimension_group timeframes.

View `separationreason` (Power BI: SeparationReason)

File: views/separationreason.view.lkml | columns: 2 | PK guess: SeparationTypeID | date cols: none

- * Confirm primary_key matches warehouse uniqueness.
- * Map dates to dimension_group timeframes.

Step C - Explores and joins

Primary explore base: FactTable. Set type + relationship + sql_on on every join.

Employee[date] -> Date[Date] [M:1, ACTIVE]

Looker action: join date relationship from PBI M:1

Employee[FP] -> FP[FP] [M:1, ACTIVE]

Looker action: join fp relationship from PBI M:1

Employee[EthnicGroup] -> Ethnicity[Ethnic Group] [M:1, ACTIVE]

Looker action: join ethnicity relationship from PBI M:1

Employee[Gender] -> Gender[ID] [M:1, ACTIVE]

Looker action: join gender relationship from PBI M:1

Employee[PayTypeID] -> PayType[PayTypeID] [M:1, ACTIVE]

Looker action: join paytype relationship from PBI M:1

Employee[BU] -> BU[BU] [M:1, ACTIVE]

Looker action: join bu relationship from PBI M:1

Employee[AgeGroupID] -> AgeGroup[AgeGroupID] [M:1, ACTIVE]

Looker action: join agegroup relationship from PBI M:1

Employee[TermReason] -> SeparationReason[SeparationTypeID] [M:1, ACTIVE]

Looker action: join separationreason relationship from PBI M:1

Step D - Measures

Mapped measures: 30. TODOs: 21.

- * [TODO] Employee.EmpCount - strategy=complex_todo
- * [TODO] Employee.Seps - strategy=complex_todo
- * [TODO] Employee.Actives - strategy=complex_todo
- * [TODO] Employee.New Hires - strategy=complex_todo
- * [TODO] Employee.AVG Tenure Days - strategy=complex_todo
- * [TODO] Employee.AVG Tenure Months - strategy=complex_todo
- * [TODO] Employee.AVG Age - strategy=complex_todo
- * [TODO] Employee.Sum of BadHires - strategy=complex_todo
- * [TODO] Employee.New Hires SPLY - strategy=complex_todo
- * [TODO] Employee.Actives SPLY - strategy=complex_todo
- * [TODO] Employee.Seps SPLY - strategy=complex_todo
- * [TODO] Employee.EmpCount SPLY - strategy=complex_todo
- * [TODO] Employee.Seps YoY Var - strategy=complex_todo
- * [TODO] Employee.Actives YoY Var - strategy=complex_todo
- * [TODO] Employee.New Hires YoY Var - strategy=complex_todo
- * [OK] Employee.Seps YoY % Change - strategy=ratio
- * [OK] Employee.Actives YoY % Change - strategy=ratio
- * [OK] Employee.New Hires YoY % Change - strategy=ratio
- * [TODO] Employee.Bad Hires SPLY - strategy=complex_todo
- * [TODO] Employee.Bad Hires YoY Var - strategy=complex_todo
- * [OK] Employee.Bad Hires YoY % Change - strategy=ratio
- * [OK] Employee.TO % - strategy=ratio
- * [TODO] Employee.TO % Norm - strategy=complex_todo
- * [TODO] Employee.TO % Var - strategy=complex_todo
- * [OK] Employee.Sep%ofActive - strategy=ratio
- * [OK] Employee.Sep%ofSMLYActives - strategy=ratio
- * [OK] Employee.BadHire%ofActives - strategy=ratio
- * [OK] Employee.BadHire%ofActiveSPLY - strategy=ratio
- * [TODO] BU.Count of BU - strategy=complex_todo
- * [TODO] Date.Count of Date - strategy=complex_todo

Step E - Calculated columns (build in warehouse)

7 calculated columns - materialize then expose as dimensions.

- * BU.Region: mid([RegionSeq], 3,15)
- * Date.MonthIncrementNumber: ([Year]-MIN([Year]))*12 +[MonthNumber]
- * Employee.isNewHire: IF(YEAR([date]) = YEAR([HireDate]) && MONTH([date])=MONTH([HireDate]), 1)
- * Employee.AgeGroupID: IF([Age]<30, 1, IF([Age]<50, 2, 3))
- * Employee.TenureDays: IF([date]-[HireDate]<0,[HireDate]-[date],[date]-[HireDate])
- * Employee.TenureMonths: CEILING([TenureDays]/30, 1) -1
- * Employee.BadHires: IF(OR((((HireDate)-[TermDate])*-1)>=61,ISBLANK([TermDate])),0,1)

Step F - Power Query M (Looker / warehouse equivalent)

Best practice: warehouse table/view + straight LookML view. SDT only as temporary bridge. Heavy M stays in ETL. Full stubs are in LOOKML_PROJECT.zip under lookml/m_migration/.

BU -> lookml_sql_derived_table

Looker object: warehouse view/table preferred; LookML SQL derived table acceptable if SELECT-only

Why: M wraps a SQL statement. Prefer creating a warehouse view/table with that SQL, then a straight

LookML view. A LookML sql-derived table is an acceptable temporary equivalent.

- * 1. Prefer warehouse VIEW/TABLE with the M SQL (or source table).
- * 2. Use straight LookML view with sql_table_name.
- * 3. Use SDT only if warehouse object is not ready yet; migrate off SDT later.
- * Files: m_migration/sql/bu.sql and lookml_stubs/bu_recommended.lkml

FP -> lookml_sql_derived_table

Looker object: warehouse view/table preferred; LookML SQL derived table acceptable if SELECT-only

Why: M wraps a SQL statement. Prefer creating a warehouse view/table with that SQL, then a straight

LookML view. A LookML sql-derived table is an acceptable temporary equivalent.

- * 1. Prefer warehouse VIEW/TABLE with the M SQL (or source table).
- * 2. Use straight LookML view with sql_table_name.
- * 3. Use SDT only if warehouse object is not ready yet; migrate off SDT later.
- * Files: m_migration/sql/fp.sql and lookml_stubs/fp_recommended.lkml

PayType -> lookml_sql_derived_table

Looker object: warehouse view/table preferred; LookML SQL derived table acceptable if SELECT-only

Why: M wraps a SQL statement. Prefer creating a warehouse view/table with that SQL, then a straight

LookML view. A LookML sql-derived table is an acceptable temporary equivalent.

- * 1. Prefer warehouse VIEW/TABLE with the M SQL (or source table).
- * 2. Use straight LookML view with sql_table_name.
- * 3. Use SDT only if warehouse object is not ready yet; migrate off SDT later.
- * Files: m_migration/sql/paytype.sql and lookml_stubs/paytype_recommended.lkml

SeparationReason -> lookml_sql_derived_table

Looker object: warehouse view/table preferred; LookML SQL derived table acceptable if SELECT-only

Why: M wraps a SQL statement. Prefer creating a warehouse view/table with that SQL, then a straight

LookML view. A LookML sql-derived table is an acceptable temporary equivalent.

- * 1. Prefer warehouse VIEW/TABLE with the M SQL (or source table).
- * 2. Use straight LookML view with sql_table_name.
- * 3. Use SDT only if warehouse object is not ready yet; migrate off SDT later.
- * Files: m_migration/sql/separationreason.sql and lookml_stubs/separationreason_recommended.lkml

Date -> lookml_sql_derived_table

Looker object: warehouse view/table preferred; LookML SQL derived table acceptable if SELECT-only

Why: M wraps a SQL statement. Prefer creating a warehouse view/table with that SQL, then a straight

LookML view. A LookML sql-derived table is an acceptable temporary equivalent.

- * 1. Prefer warehouse VIEW/TABLE with the M SQL (or source table).
- * 2. Use straight LookML view with sql_table_name.
- * 3. Use SDT only if warehouse object is not ready yet; migrate off SDT later.
- * Files: m_migration/sql/date.sql and lookml_stubs/date_recommended.lkml

Employee -> warehouse_transform_model

Looker object: warehouse transform + straight view (avoid SDT for heavy M)

Why: M contains merge, append/union, or heavy transforms. Looker skills: keep LookML semantic; put ETL in the warehouse.

- * 1. Read phase1/inventory/04_m_raw/Employee.m end-to-end.
- * 2. Implement joins/unions/filters in dbt/Dataform/SQL.
- * 3. Expose curated table to LookML views/employee.view.lkml via sql_table_name.
- * 4. Mark any LookML SDT as temporary technical debt.
- * Files: m_migration/sql/employee.sql and lookml_stubs/employee_recommended.lkml

Ethnicity -> warehouse_seed_plus_straight_view

Looker object: warehouse seed table + straight view (tiny SDT optional)

Why: Embedded/static M tables should become warehouse seeds. Straight LookML view afterward; SDT only for tiny temporary seeds.

- * 1. Extract static rows from M into a seed CSV or INSERT script.
- * 2. Load seed to warehouse.
- * 3. Use generated straight LookML view.
- * Files: m_migration/sql/ethnicity.sql and lookml_stubs/ethnicity_recommended.lkml

Gender -> warehouse_seed_plus_straight_view

Looker object: warehouse seed table + straight view (tiny SDT optional)

Why: Embedded/static M tables should become warehouse seeds. Straight LookML view afterward; SDT only for tiny temporary seeds.

- * 1. Extract static rows from M into a seed CSV or INSERT script.
- * 2. Load seed to warehouse.
- * 3. Use generated straight LookML view.
- * Files: m_migration/sql/gender.sql and lookml_stubs/gender_recommended.lkml

AgeGroup -> warehouse_seed_plus_straight_view

Looker object: warehouse seed table + straight view (tiny SDT optional)

Why: Embedded/static M tables should become warehouse seeds. Straight LookML view afterward; SDT only for tiny temporary seeds.

- * 1. Extract static rows from M into a seed CSV or INSERT script.
- * 2. Load seed to warehouse.
- * 3. Use generated straight LookML view.
- * Files: m_migration/sql/agegroup.sql and lookml_stubs/agegroup_recommended.lkml

6. Gaps - prioritize and close

Each gap blocks production readiness until resolved or formally accepted.

Severity HIGH (4)

GAP [HIGH] - Connection & warehouse

What is missing: LookML still uses placeholders YOUR_LOOKER_CONNECTION and YOUR_PROJECT.YOUR_DATASET.

What to do:

Set the real Looker connection name and point every sql_table_name at existing warehouse tables before validating.

GAP [HIGH] - Power Query / ETL

What is missing: 9 Power Query queries need warehouse/Looker equivalents. Pattern mix:

```
{'lookml_sql_derived_table': 5, 'warehouse_transform_model': 1, 'warehouse_seed_plus_straight_view': 3}.
```

What to do:

Open LOOKML_PROJECT.zip -> lookml/m_migration/. For each query follow M_QUERY_RECOMMENDATIONS.md, implement sql/<query>.sql in the warehouse, then update the straight LookML view sql_table_name. Use LookML SDT only when the recommendation allows a temporary bridge.

GAP [HIGH] - Complex DAX

What is missing: 21 of 30 measures are TODO stubs (CALCULATE/time-intel/iterators).

What to do:

Implement each TODO using LookML filters, period-over-period patterns, or warehouse logic. Keep original DAX in the field description until KPI parity passes.

GAP [HIGH] - Calculated columns

What is missing: 7 business calculated columns must be materialized (prefer warehouse), not left as DAX.

What to do:

Create warehouse columns for: BU.Region; Date.MonthIncrementNumber; Employee.isNewHire; Employee.AgeGroupID; Employee.TenureDays; Employee.TenureMonths; Employee.BadHires

Severity LOW (2)

GAP [LOW] - Row-level security

What is missing: No RLS roles in this PBIX inventory.

What to do:

Still confirm with security owners whether Looker needs access_filter by region/org.

GAP [LOW] - Auto date tables

What is missing: 6 LocalDateTable_/DateTableTemplate_ tables skipped (correct).

What to do:

Use business date columns with dimension_group timeframes. Do not migrate auto-date tables.

7. What to check (validation)

LookML compile

- * ☐ Validator: 0 errors
- * ☐ One primary_key per view
- * ☐ Every join has relationship + sql_on
- * ☐ Dialect-correct SQL

Grain and joins

- * ☐ Fact row count matches Power BI (same filters)
- * ☐ count vs count_distinct(PK) shows no unexpected fan-out
- * ☐ Orphan FK rate understood for left_outer joins

Business logic

- * ☐ All HIGH gaps closed or signed off
- * ☐ Calculated columns available
- * ☐ Required measures implemented and spot-checked vs Power BI
- * ☐ Inactive / M:M joins reviewed

Security and ops

- * ☐ RLS / access_filter decision recorded
- * ☐ Caching / datagroup set
- * ☐ QA explores hidden from end users

8. Acceptance criteria

- * Validator clean on target connection
- * HIGH gaps closed or formally accepted
- * At least 5 Power BI business questions reproduce in Looker within agreed tolerance
- * Remaining MEDIUM/LOW gaps have owners and dates
- * Do NOT publish claiming full KPI parity while DAX TODOs remain

9. Inventory snapshot

- * tables: 15
- * business_tables: 9
- * internal_tables: 6
- * columns: 87
- * measures: 30
- * calculated_columns: 43
- * calculated_tables: 6
- * relationships: 8
- * power_query: 9
- * m_files: 9
- * rls_roles: 0
- * hierarchies: 7
- * partitions: 122
- * auto_date_tables: 6
- * annotations: 158
- * sort_by_columns: 0
- * format_strings: 5
- * perspectives: 0
- * display_folders: 0

10. Out of scope

- * Power BI report pages, visuals, bookmarks, themes
- * Automatic warehouse DDL generation
- * Automatic KPI certification

Generated by phase2/generate_developer_guide.py for Human Resources Sample PBIX.pbix.